

Sales Analytics Pipeline Project

1. Project Overview

The objective of this project is to design and implement an end-to-end data engineering pipeline that provides a unified and analytical view of inbound sales performance.

The system should:

- Track the complete inbound sales funnel: Triage Call → Strategy Call → Outcome
- Measure conversion efficiency across each stage of the funnel
- Evaluate sales team performance (setters/closers)
- Identify drop-offs and inefficiencies in the sales process
- Provide time-based insights (daily/weekly trends)
- Deliver insights through a BI dashboard for business stakeholders

The final solution should support data-driven decision-making for sales operations and performance optimisation.

2. Source Data Explanation

Task

- Extract data from PostgreSQL source tables
- Understand schema, relationships, and data characteristics
- Identify key entities required for building the funnel

Source System Details

Database: PostgreSQL

Connection Details:

Parameters	Values	Schema	Tables
host_name	http://dea.cgyi97rb4alr.us-east-1.rds.amazonaws.com/	raw	leads_raw
db_name	dea_analytics_dev	raw	lead_activites_raw
db_user	student_user	raw	close_crm_users_raw
db_password	DataEngineer12345	raw	custom_activites_raw

db_port	5432		
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Source data is updated daily by 07:30 AM EST.

2.1 Lead Activities Dataset (Raw Events)

- *Includes events:*
 - Custom Activities (Triage, Strategy, Sale)
 - SMS events
 - Calls (outbound/inbound)
 - Meetings
 - Notes
- Delivered as **nested JSON** under the key:
`masked_activities: [ { ...activity object... } ]`

Activity types:

Type	Meaning
CustomActivity	Triage call, strategy call, sale outcome
SMS	Automated or manual messages
Call	Dialed or received calls
Meeting	Strategy call calendar event
Note	Sales notes

2.2 Custom Activities (Dimension Data)

Metadata describing *what each custom activity type means*:

- Custom activity type ID → e.g. `actitype_38341SW0KRkRHHAqWEqSJu`
- Field IDs → e.g. `cf_tHx0wx1Ysk0Xo6sRXhqIHToin8itnyaCHC1vkmHqugh`
- Field labels → "Triage Call Date"
- Data types → text, date, choice, user, etc.
- Choice values → e.g.
 - 1. Strategy Call Booked
 - 6. No Show

2.3 Close CRM Users (Dimension Data)

- User ID → e.g., `user_rV7fyf1LH4c2RP6glawdjDcoS5hYH2fgsk82SFQiiJV` (A unique identifier generated for each user).
- Field IDs (Keys) → e.g., `email`, `first_name`, `last_used_timezone`, `organizations`, `date_created`.
- Field labels (Context) → "Email Address", "First Name", "Last Known Timezone", "Assigned Organizations", "Account Creation Date".
- Data types → string (text), timestamp (date/time), array (lists), null (empty/unassigned).
- Choice values / System Formats → e.g.,
 1. Timezones (Standard IANA format): `Europe/London`, `America/New_York`, `Asia/Calcutta`, `Africa/Johannesburg`.
 2. Organization IDs (Relational data):
`orga_rq7EW7TTwnrAshSzIQ49jJUWzDxtTQUBBrA4VoRWE1D` (Links the user to a specific company/workspace).
 3. Timestamps (ISO 8601 format): `2025-06-20T21:17:11.527000+00:00` (Used for `date_created`, `date_updated`, and `email_verified_at`).

Important Data Property

Each daily data refresh contains:

- **All historical activities** for leads updated on that day
- Meaning: **same activity may appear on multiple days**
- This introduces **duplicates** → pipeline must deduplicate.
- **The stringified data within `JSON_OBJECT`** contains: **Single quotes (')** enclosing properties and values instead of standard **double quotes (")**.

- Meaning: **Strict standard JSON parsers will throw a syntax error** when attempting to read this inner string.
- **This causes parsing failures** → pipeline must sanitize/replace single quotes with double quotes before executing the second parse.

3. Understanding the Business Hierarchy

In this project, we deal with lead activities captured at different stages in the sales funnel. Leads generally follow two distinct paths depending on how they are acquired: Inbound or Outbound. Every lead goes through multiple steps (or may drop off at some point).

The key stages are:

- **Initial Contact:** Triage Call Activity (Inbound) or Prospecting Activity (Outbound) – This is the first step.
- **Strategy Call Activity:** This occurs if the initial contact outcome is positive.
- **Sales Outcome:** This is the final step if the strategy call converts into a sale.

Note: While SMS, regular Calls, and Emails are present in the data, the core KPIs are driven by Custom Activities reflecting the funnel stages above.

3.1 Initial Contact (Inbound Triage & Outbound Prospecting)

Logged as a CustomActivity in the lead activity data. Represents the initial qualification effort.

Key outcomes:

- **Strategy Call Scheduled:** Lead qualifies for the next step. Mapped specifically as '1. Strategy Call Scheduled' for inbound and '2. Strategy Call Scheduled' for outbound.
- **No Show/Cancel/Reschedule:** Lead didn't attend or requires rescheduling (e.g., '6. No Show', '8. Cancel', '7. Reschedule').

3.2 Strategy Call Activity

If the initial contact results in a scheduled strategy call. This call is more in-depth and typically handled by a senior salesperson or closer. Logged under activities like '5) Strategy Call' or '6) Strategy Call Follow Up'.

Key outcomes:

- **Sale:** Lead buys (e.g., '5. Sale', '6. Sale').

- **Lost/Cancel:** No sale or lead dropped (e.g., '7. Lost', '2. Admin Cancel', '3. Cancel-Not Interested').
- **Follow Up:** Needs another meeting (e.g., '1. Follow Up').

3.3 Sales Outcome

Captures final sale data, derived from specific sales activities like '7) New Sale' or '8) New Sale [Custom Payment Plan]'.

Key information:

- **Sale Status:** Completed.
- **Contracted Value:** Total deal size.
- **Cash Collected:** Immediate revenue captured.

3.4 Putting it Together: Full Lead Journey Example

Lead A (Success Case)

1. **Triage Call** (July 1st)
 - Outcome: Strategy Call Booked
2. **Strategy Call** (July 2nd)
 - Outcome: Sale
3. **Sale Recorded**
 - Contracted Value: \$15,000

Lead A counts as: **Inbound Booked, Inbound Taken, Strategy Call Booked, Sale**

Lead B (Drop-Off Case)

1. **Triage Call** (July 1st)
 - Outcome: No Show

Lead B counts as: **Inbound Booked**, but **not Inbound Taken** and no strategy call or sale.

Lead C (No Sale Case)

1. **Triage Call** (July 1st)
 - Outcome: Strategy Call Booked
2. **Strategy Call** (July 2nd)
 - Outcome: Lost

Lead C counts as: **Inbound Booked, Inbound Taken, Strategy Call Booked**, but **no sale**.

Why is this Hierarchy Important?

- All KPIs (Show Rate, Sale Rate, Offer Rate) are derived by **linking activities in sequence**.
- If we don't respect the hierarchy, we might **double-count activities** or misclassify leads.

4. Data Processing Requirements

4.1 Deduplication & Cleansing Logic

Each daily batch contains historical activities, leading to potential duplicates. The data also contains malformed JSON strings that must be repaired before parsing.

Required approach:

- **Data Repair:** Implement a custom UDF to repair malformed strings, neutralize messy text fields (like descriptions or notes), and escape apostrophes prior to JSON parsing.
- **Deep Unwrapping:** Dynamically unwrap nested arrays (`raw_data`, `data`) to access the base activity objects.
- **Change Data Capture (CDC):** Identify duplicates and track historical changes using an `MD5_HASH` generated from concatenating key row attributes (e.g., `lead_id`, `activity_at`, `status_change`, etc.).
- **Upserts:** Use `MERGE INTO` statements to insert new records and update existing ones only when the `MD5_HASH` changes, utilizing `INSERT_DATE` and `UPDATE_DATE` for auditability.
- **Identify duplicates using:** `lead_id` + `activity_id`
- Keep the **latest activity record** using: `activity_at` (activity timestamp)
- Use **upserts** (insert new, update existing) into warehouse tables.

5. Phased Project Execution

Phase 1: Architecture Design (Bronze → Silver → Gold)

You must propose:

- How to ingest JSON from S3 into Bronze tables using Snowflake COPY INTO.
- How to flatten nested structures, repair JSON, and calculate MD5 hashes for the Silver layer.
- How to design Dimensions & Facts (Gold Views) to serve the 4 specific reporting models.

Phase 2: Data Modeling Students design complete ERD including:

- **Bronze tables:** `CLOSE_CRM_USERS_RAW`, `CUSTOM_ACTIVITIES_RAW`, `LEAD_ACTIVITIES_RAW`.
- **Silver & Gold layer entities to be designed.**

Phase 3: Development & Deployment You must execute the following:

- **Ingestion:** Create Stages and COPY INTO Bronze tables.
- **Processing:** Normalize arrays into Silver tables. Map dynamic activity types by extracting keys containing `%custom.%` and joining them to the customized outcome dictionary.
- **Build Funnel Logic:** Link initial contact → strategy → sale per lead using timestamps and activity outcomes to identify the correct sequence.
- **Orchestration:** Schedule daily job runs. Track incremental loads and purge processed files from stages using dynamic date patterns.
- Handles **schema evolution** and nested custom activity structures
- **Implements deduplication** because every daily batch contains historical data
- **Produces a KPI report** summarizing inbound lead performance
- **Runs daily for 7 days** (simulating a production pipeline)
- **The pipeline must use a Bronze → Silver → Gold medallion architecture.**

The end to end architecture and data modeling should be verified with the SME before proceeding with implementation.

6. Business KPIs & Final Reports

The Gold layer must power four distinct KPI reports:

6.1 Inbound Setter Report

Tracks the performance of setters handling inbound triage calls.

- **Inbound Booked & Taken:** Total triage calls logged, and those where the lead actually showed up.
- **Show Rate:** % of booked inbound calls taken.

- **Triage Set Rate:** % of inbound calls resulting in a strategy call being scheduled.
- **Strategy Call Booked & Taken:** Volume of follow-up strategy calls set and attended (excluding cancellations and no-shows).
- **Offer Rate:** % of taken strategy calls where an offer was explicitly presented.
- **Total Sales & Sale Rate:** Total closed deals and the % of taken inbound calls that resulted in a sale.
- **Average Order Value:** Total revenue divided by total sales.

6.2 Outbound Setter Report

Tracks the performance of setters making outbound prospecting dials.

- **Total Outbound Calls & Leads Touched:** Raw dial volume and unique leads contacted.
- **Outbound Set:** Calls that successfully scheduled a Strategy Call.
- **Total Closer Show:** Strategy calls where the prospect actually attended.
- **Dial to Set Rate:** % of raw dials that convert to a booked meeting.
- **Set to Show Rate:** % of scheduled outbound meetings that occur.
- **Show to Sale Rate:** % of attended outbound meetings that close.

6.3 Closer Report

Tracks the performance of the sales closers handling strategy calls.

- **Call Booked:** Total strategy calls assigned to the closer.
- **Cancellation Breakdown:** Volume of calls canceled by admin, canceled for nurture, or canceled due to lack of interest.
- **No Show vs. Show:** Volume of leads who missed the call versus those who attended.
- **Lost vs. Sale:** Volume of attended calls that did not close versus those that generated a sale.
- **Financials:** Average contract value and total cash collected per closer.

6.4 Objections Faced Report

Analyzes the reasons prospects do not buy during the Strategy Call.

- **Total Calls:** Unique strategy calls evaluated.
- **Objection Categories (Count & %):** Tracks the volume and percentage of calls blocked by: Money, Fear, Hung Up, Logistical, No Objections, Talking to Other Coaches, Partner, Think About It, Time, Trust, Value, and Wasn't Looking For What We Offered.